Dr. Guth Data Examination Report

Project: Dr. Guth Data Examination
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1 Introduction

Dr. Guth's study looks at whether pre-clerkship OSCE scores differ between students who are underrepresented in medicine and those who are not at the University of Colorado School of Medicine. Clerkship grade is of interest due to its link to previous literature, this report focuses on OPC final course grade. Before answering the research question,the data needed to be checked for missingness, incorrect coding, and unusual values. This report describes those checks and changes made through R markdown, and the descriptive results after cleaning. No comparisons between groups were conducted.

2 Methods

I imported the dataset and coding guide into R and created a separate copy for cleaning. The original Data is unchanged. I checked the number of records and variables, missing and repeated student IDs, duplicate rows, missing values, and blank entries. I also checked selected categorical values against the coding guide and examined the ranges of USMLE and OSCE scores. Blank text entries were changed to NA, and categorical codes were given readable labels. Gender code "D" was labeled Non-binary following clarification from zoom meeting, and unknown URM status was kept as a separate category. Underrepresented in medicine status is defined using the school's own URM variable, which reflects groups considered underrepresented at the University of Colorado School of Medicine. The AAMC classification (URM_AAMC), which is based on underrepresentation nationally, is not used as the grouping variable. No student was removed, and missing value was not replaced. I used box plots to identify potentially unusual OSCE scores and histograms to examine the shape and spread of each OSCE score distribution. Histogram intervals were set at five percentage points. Unusual scores were retained unless there was evidence that they were incorrect. Numerical variables were summarized using the number of observed scores, mean, standard deviation, median, quartiles, and range. Missing scores were excluded separately for each numerical summary and plot. Categorical variables were summarized using counts and percentages out of all 199 students, with missing and unknown categories shown separately. Report focused on OSCE scores, both URM, gender, USMLE measures, and OPC final course grade. No comparisons between groups and no hypothesis tests.

2.1 Data Checking and Cleaning

```
## [1] 199
```

```
## [1] 52
```

```
## [1] 0
```

```
## [1] 0
```

```
## [1] 0
```

Total Number of Records is 199 and 52 variables. Based on the check, no records were removed.

Table 1: Missing and blank entries.

Variable	Missing-NA	Blank	Missing or blank (%)
AAC_clinical_grade	180	0	90.5
CPC_clinical_grade	66	0	33.2
HAC_clinical_grade	23	0	11.6
ICAC_clinical_grade	28	0	14.1
NC_clinical_grade	38	0	19.1
OBG_clinical_grade	49	0	24.6
OPC_clinical_grade	41	0	20.6
PC_clinical_grade	47	0	23.6
RCC_clinical_grade	180	0	90.5
AAC_Final_Course_Grade	179	0	89.9
CPC_Final_Course_Grade	46	0	23.1
HAC_Final_Course_Grade	20	0	10.1
ICAC_Final_Course_Grade	27	0	13.6
MSK_Final_Course_Grade	34	0	17.1
NC_Final_Course_Grade	7	0	3.5
OBG_Final_Course_Grade	28	0	14.1
OPC_Final_Course_Grade	20	0	10.1
PC_Final_Course_Grade	26	0	13.1
RCC_Final_Course_Grade	179	0	89.9
OPC_Clin_Grade_Sub	61	0	30.7
OPC_Clin_Grade_Anes	47	0	23.6
STEP2CK_PassFail	0	1	0.5
STEP2CK_ExamScore	1	0	0.5
amcasEthnicityDesc	0	161	80.9
amcas_othethnicity	199	0	100.0
additional_ethnicity	0	2	1.0
race_desc	0	23	11.6
race_cat_desc	0	23	11.6
additional_race	0	51	25.6
P1SprBates_PhysExamScore	42	0	21.1
P1SprBates_CommScore	42	0	21.1
P1SprBates_MedDocScore	42	0	21.1
P1FallBodySys_PhysExamScore	46	0	23.1
P2FallRamos_PhysExamScore	41	0	20.6
P2FallRamos_CommScore	41	0	20.6
P2FallRamos_OralPresScore	41	0	20.6
P2FallPayne_PhysExamScore	41	0	20.6
P2FallPayne_CommScore	41	0	20.6
P2FallPayne_MedDocScore	41	0	20.6
P2SpringHyde_PhysExamScore	42	0	21.1
P2SpringHyde_CommScore	42	0	21.1
P2SpringHyde_MedDocScore	42	0	21.1
P2SpringSnelling_PhysExamScore	42	0	21.1
P2SpringSnelling_CommScore	42	0	21.1

Variable	Missing-NA	Blank	Missing or blank (%)
P2SpringSnelling_MedDocScore	65	0	32.7

Missing and blank entries were found across several variables. OPC final grade was missing for 20 students (10.1%), and OSCE scores had 20.6%–32.7% missingness.

Table 2: Completeness of OSCE scores across students.

Category	Students	Percent
All OSCE scores observed	129	64.8
Some OSCE scores missing	31	15.6
All OSCE scores missing	39	19.6

All OSCE scores were available for 129 students (64.8%). Some scores were missing for 31 students (15.6%), and all scores were missing for 39 students (19.6%). No students were removed.

Blank entries were converted to `NA` in the working dataset to standardize the representation of missing information.

Table 3: Frequency distribution: OPC_Final_Course_Grade

Value	Students
1	41
2	66
3	72
NA	20

Table 4: Frequency distribution: gender

Value	Students
D	1
F	101
M	97

Table 5: Frequency distribution: URM

Value	Students
0	132
1	60
3	7

Table 6: Frequency distribution: URM_AAMC

Value	Students
0	140
1	52

Value	Students
3	7

Table 7: Frequency distribution: STEP1_PassFail

Value	Students
F	12
P	187

Table 8: Frequency distribution: STEP2CK_PassFail

Value	Students
P	198
NA	1

The selected categorical variables contained expected codes. URM code 3 represented Unknown, and gender code "D" was confirmed as Nonbinary.

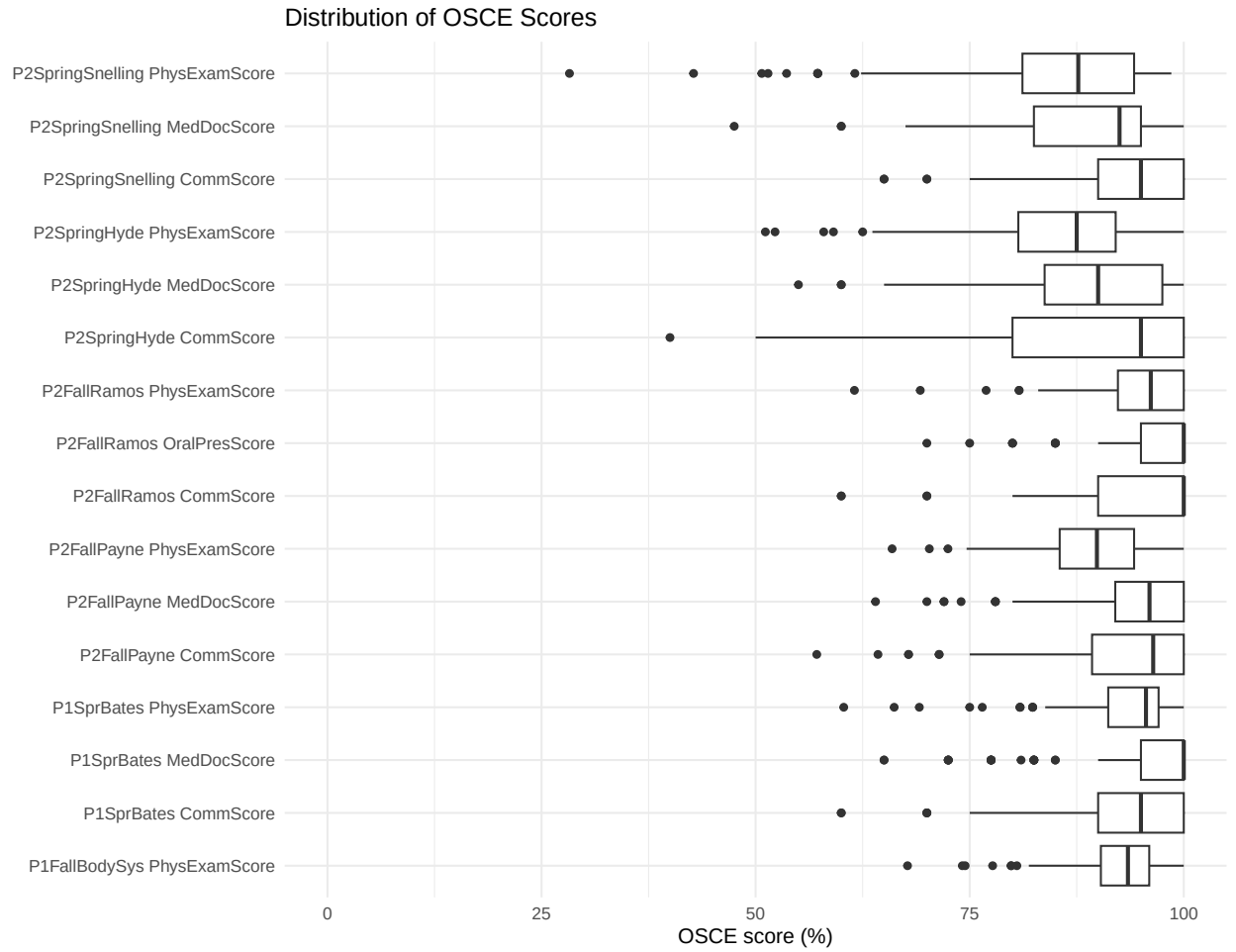
Categorical codes were converted to readable labels. Gender code "D" was labeled "Nonbinary" following clarification. Unknown URM status was retained as a separate category.

Table 9: Observed ranges of USMLE and OSCE scores.

Variable	Observed	Minimum	Maximum
STEP1_ExamScore	199	164.00	267.00
STEP2CK_ExamScore	198	209.00	276.00
P1SprBates_PhysExamScore	157	60.29	100.00
P1SprBates_CommScore	157	60.00	100.00
P1SprBates_MedDocScore	157	65.00	100.00
P1FallBodySys_PhysExamScore	153	67.74	100.00
P2FallRamos_PhysExamScore	158	61.54	100.00
P2FallRamos_CommScore	158	60.00	100.00
P2FallRamos_OralPresScore	158	70.00	100.00
P2FallPayne_PhysExamScore	158	65.94	100.00
P2FallPayne_CommScore	158	57.14	100.00
P2FallPayne_MedDocScore	158	64.00	100.00
P2SpringHyde_PhysExamScore	157	51.14	100.00
P2SpringHyde_CommScore	157	40.00	100.00
P2SpringHyde_MedDocScore	157	55.00	100.00
P2SpringSnelling_PhysExamScore	157	28.26	98.55
P2SpringSnelling_CommScore	157	65.00	100.00
P2SpringSnelling_MedDocScore	134	47.50	100.00

```
## [1] 0
```

Step 1 scores ranged from 164 to 267 among 199 students. Step 2 CK scores ranged from 209 to 276 among 198 students. All observed OSCE scores were within 0–100%. The lowest OSCE score was 28.26% on the P2 Spring Snelling physical examination assessment.



Most OSCE scores were concentrated near 100%, with several low scores beyond the box plot whiskers.

2.2 Descriptive Statistics

Table 10: Descriptive statistics for observed USMLE and OSCE scores.

Variable	Observed	Mean	SD	Median	Q1	Q3
STEP1_ExamScore	199	226.34	20.02	227.00	211.50	242.00
STEP2CK_ExamScore	198	243.03	13.87	243.50	233.00	254.00
P1SprBates_PhysExamScore	157	93.34	6.39	95.59	91.18	97.06
P1SprBates_CommScore	157	91.59	8.44	95.00	90.00	100.00
P1SprBates_MedDocScore	157	95.66	7.56	100.00	95.00	100.00
P1FallBodySys_PhysExamScore	153	92.42	5.34	93.48	90.32	95.97
P2FallRamos_PhysExamScore	158	94.94	6.24	96.15	92.31	100.00
P2FallRamos_CommScore	158	94.81	7.53	100.00	90.00	100.00
P2FallRamos_OralPresScore	158	96.87	5.45	100.00	95.00	100.00
P2FallPayne_PhysExamScore	158	88.87	6.22	89.86	85.51	94.20
P2FallPayne_CommScore	158	93.19	8.08	96.43	89.29	100.00
P2FallPayne_MedDocScore	158	93.54	7.25	96.00	92.00	100.00
P2SpringHyde_PhysExamScore	157	85.34	9.64	87.50	80.68	92.05
P2SpringHyde_CommScore	157	89.71	12.16	95.00	80.00	100.00

Variable	Observed	Mean	SD	Median	Q1	Q3
P2SpringHyde_MedDocScore	157	88.17	10.15	90.00	83.75	97.50
P2SpringSnelling_PhysExamScore	157	85.30	11.98	87.68	81.16	94.20
P2SpringSnelling_CommScore	157	92.64	8.46	95.00	90.00	100.00
P2SpringSnelling_MedDocScore	134	89.56	9.85	92.50	82.50	95.00

Among students with observed scores, the mean Step 1 score was 226.34 (SD = 20.02), and the mean Step 2 CK score was 243.03 (SD = 13.87). Across OSCE components, mean scores ranged from 85.30% to 96.87%, with 134–158 observations available per component. Several components had medians of 100%, suggesting scores were concentrated near the upper limit. These distributions support reporting medians and quartiles alongside means and standard deviations.

Table 11: Categorical summaries after cleaning. Percentages use all 199 students.

Variable	Category	Students	Percent
OPC_Final_Course_Grade	Fail	0	0.0
OPC_Final_Course_Grade	Pass	41	20.6
OPC_Final_Course_Grade	High Pass	66	33.2
OPC_Final_Course_Grade	Honors	72	36.2
OPC_Final_Course_Grade	Pass with remediation	0	0.0
OPC_Final_Course_Grade	Missing	20	10.1
gender	Female	101	50.8
gender	Male	97	48.7
gender	Nonbinary	1	0.5
URM	Non-URM	132	66.3
URM	URM	60	30.2
URM	Unknown	7	3.5
URM_AAMC	Non-URiM	140	70.4
URM_AAMC	URiM	52	26.1
URM_AAMC	Unknown	7	3.5
STEP1_PassFail	Fail	12	6.0
STEP1_PassFail	Pass	187	94.0
STEP2CK_PassFail	Fail	0	0.0
STEP2CK_PassFail	Pass	198	99.5
STEP2CK_PassFail	Missing	1	0.5

Of the 199 students, 101 (50.8%) were female, 97 (48.7%) were male, and one (0.5%) was nonbinary. Institution-specific URM status identified 60 students (30.2%) as underrepresented, compared with 52 (26.1%) using the AAMC classification; each variable recorded seven students (3.5%) as unknown. OPC final grades were Pass for 41 students (20.6%), High Pass for 66 (33.2%), and Honors for 72 (36.2%), with 20 grades missing (10.1%). Step 1 status indicated 187 passes (94.0%) and 12 failures (6.0%), while Step 2 CK status indicated 198 passes (99.5%) and one missing entry (0.5%). All percentages use the full sample of 199

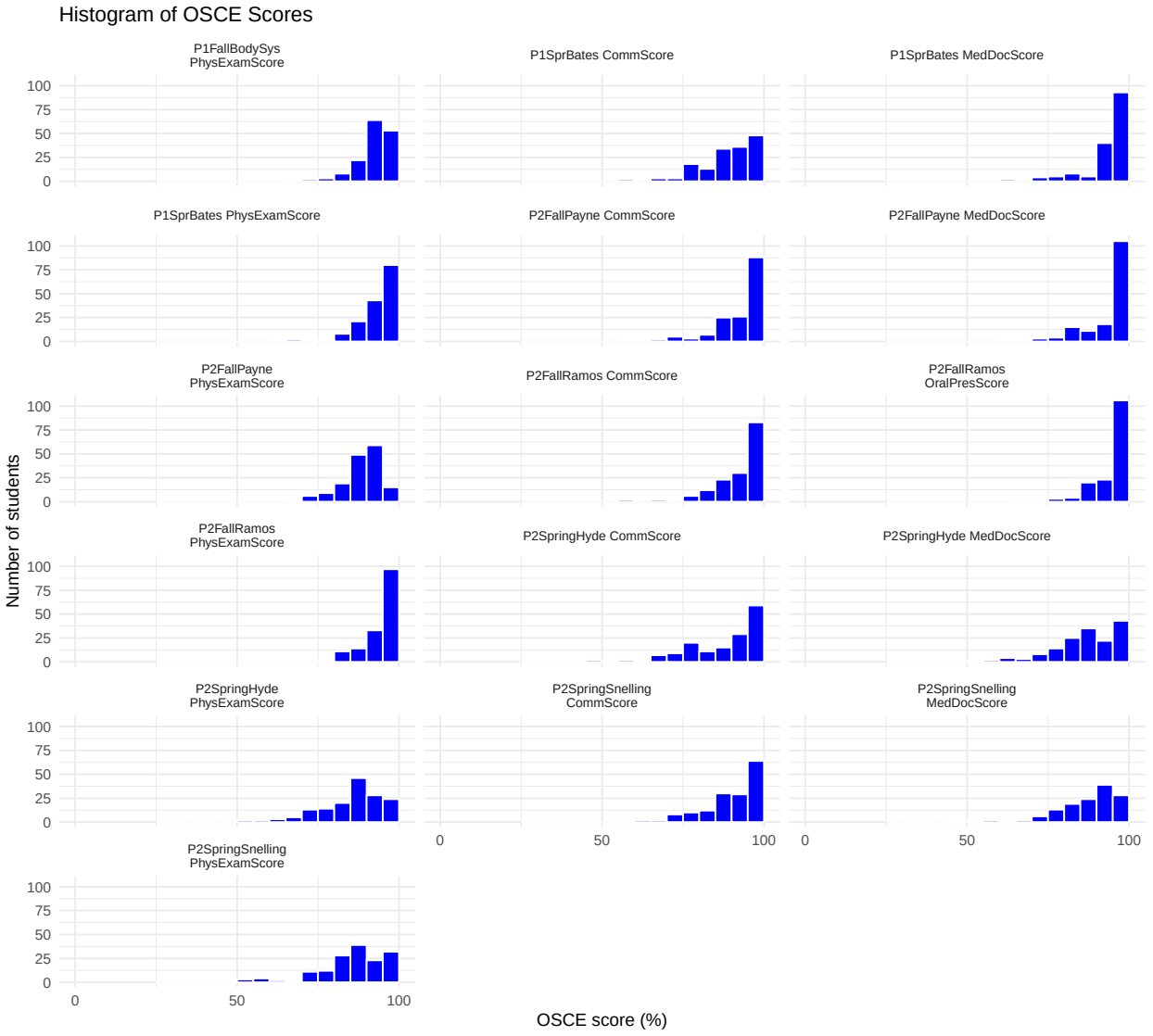


Figure 1: Distributions of observed OSCE scores. Missing scores are excluded.

3 Conclusions

Missing information was the main concern with 39 students who are missing all the OSCE scores. THere are no missing or repeated IDs, duplicate rows, or unexpected codes in the selected categorical variables. All observed OSCE scores were within 0%–100%. Blank entries were changed to NA, and categorical codes were labeled. All 199 students were retained. Reasons for missing scores, unusually low scores. There is also the difference between the expected sample of 180, but dataset contains 199 records, may need clearification.

4 AI Assistance

I used ChatGPT to develop and troubleshoot some of the the code, and to properly Knit.